EXPERIMENT-21:

AIM:

To write a C program to demonstrate ECB, CBC, and CFB modes and understand the need for padding in block cipher encryption.

CODE:

```
#include <stdio.h>

#include <string.h>


#define MAX 100


int main() {

    char p[MAX], e[MAX];

    int i, n, key = 5, iv = 7, prev, block = 4;


    printf("Enter plaintext: ");

    scanf("%99s", p);


    n = strlen(p);


    // Padding: 1 followed by zeros

    p[n++] = '1';

    while(n % block != 0)

        p[n++] = '0';

    p[n] = '\0';


    printf("\nPadded Plaintext: %s\n", p);
```

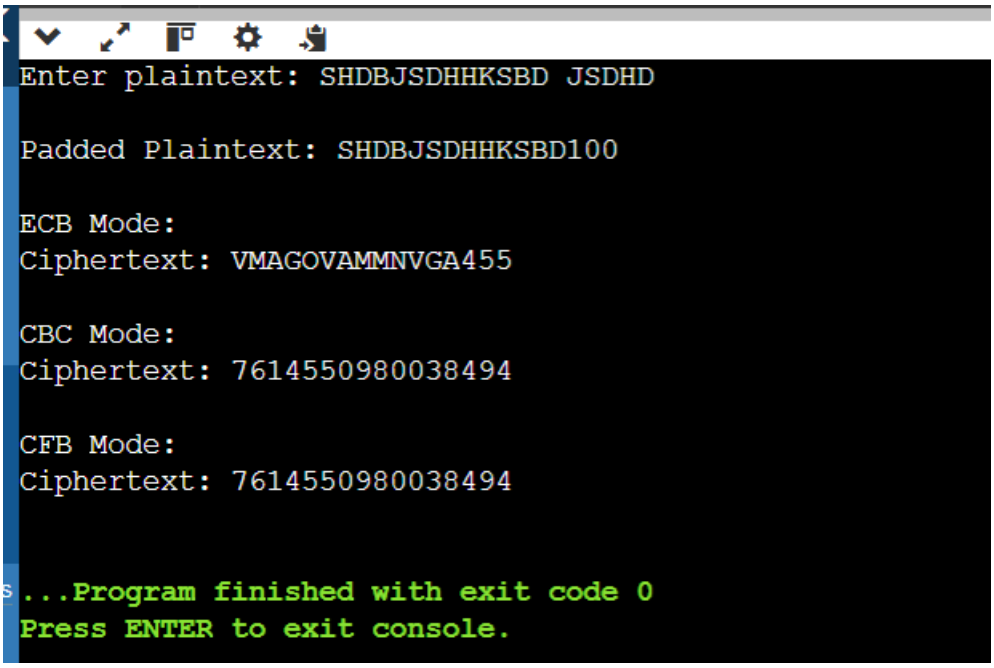
```
    printf("\nECB Mode:\n");
    for(i = 0; i < n; i++)
        e[i] = p[i] ^ key;
    e[n] = '\0';
    printf("Ciphertext: %s\n", e);

    printf("\nCBC Mode:\n");
    prev = iv;
    for(i = 0; i < n; i++) {
        prev = (p[i] - '0' + prev + key) % 10;
        e[i] = prev + '0';
    }
    e[n] = '\0';
    printf("Ciphertext: %s\n", e);

    printf("\nCFB Mode:\n");
    prev = iv;
    for(i = 0; i < n; i++) {
        e[i] = ((p[i] - '0' + prev + key) % 10) + '0';
        prev = e[i] - '0';
    }
    e[n] = '\0';
    printf("Ciphertext: %s\n", e);

    return 0;
}
```

OUTPUT:


Enter plaintext: SHDBJSDHHKSBD JSDHD
Padded Plaintext: SHDBJSDHHKSBD100
ECB Mode:
Ciphertext: VMAGOVAMMNVGA455
CBC Mode:
Ciphertext: 7614550980038494
CFB Mode:
Ciphertext: 7614550980038494
...Program finished with exit code 0
Press ENTER to exit console.